

Dispersion or Scatterness

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Ex. (i)	7, 8, 9, 10, 11	$\bar{x}_1 = 9$	$R_1 = 4$
(ii)	3, 6, 9, 12, 15	$\bar{x}_2 = 9$	$R_2 = 12$
(iii)	1, 5, 9, 13, 17	$\bar{x}_3 = 9$	$R_3 = 16$

Measures of Dispersion

Crude measures

i) Range = $H - L$ (based on only two observations)

ii) Quartile deviation or semi-inter quartile range
 $Q = \frac{1}{2}(Q_3 - Q_1)$ (Use 50% of the data ignores 50%.)

Mean Absolute Deviation from the average A (mean, median or mode) for X/f is given by

$$MADA = \frac{1}{N} \sum_{i=1}^n f_i |x_i - A| \quad ; \quad N = \sum_{i=1}^n f_i$$

A can be $\bar{x}$ (mean), Md (median) or Mo (mode)

$MADA$ is least when $A = Md$.

Drawbacks: better than range or QD but it ignores the sign of deviations $(x_i - A)$ creates artificiality and makes it useless for further mathematical treatments.

Root mean square deviation in X/f about a constant A

$$S = \sqrt{\frac{1}{N} \sum_{i=1}^n f_i (x_i - A)^2}$$

~~As we know~~ The mean square deviation is defined as

$$S^2 = \frac{1}{N} \sum_{i=1}^n f_i (x_i - A)^2$$

We know $\sum_{i=1}^n f_i (x_i - A)^2$ will be minimum iff $A = \bar{x}$

Hence, if we replace A by $\bar{x}$, we get

$$\text{Standard deviation } (\sigma) = \sqrt{\frac{1}{N} \sum_{i=1}^n f_i (x_i - \bar{x})^2}$$

$$\text{and variance } (\sigma^2) = \frac{1}{N} \sum_{i=1}^n f_i (x_i - \bar{x})^2$$

S.D. (σ) / variance (σ^2) is an ideal measure of dispersion.

For numerical the formula of variance can be simplified as

$$\begin{aligned} \sigma^2 &= \frac{1}{N} \sum_{i=1}^n f_i (x_i - \bar{x})^2 = \frac{1}{N} \sum_{i=1}^n f_i (x_i^2 + \bar{x}^2 - 2x_i \bar{x}) \\ &= \frac{1}{N} \sum_{i=1}^n f_i x_i^2 + \bar{x}^2 \sum_{i=1}^n f_i - 2\bar{x} \cdot \frac{1}{N} \sum_{i=1}^n f_i x_i \\ &= \frac{1}{N} \sum_{i=1}^n f_i x_i^2 + \bar{x}^2 - 2\bar{x}^2 \\ &= \frac{1}{N} \sum_{i=1}^n f_i x_i^2 - \bar{x}^2 \end{aligned}$$

$$\boxed{\sigma^2 = \frac{1}{N} \sum_{i=1}^n f_i x_i^2 - \left(\frac{1}{N} \sum_{i=1}^n f_i x_i \right)^2}$$

Effect of change of origin & scale on variance x/f

$$\sigma_x^2 = \frac{1}{N} \sum_{i=1}^n f_i (x_i - \bar{x})^2$$

$U = \frac{x-A}{h}$; where A & h are suitably chosen constants.

$$x = A + hU$$

$$x_i = A + hU_i$$

$$\bar{x} = A + h\bar{u} \quad , \quad x_i - \bar{x} = h(U_i - \bar{u})$$

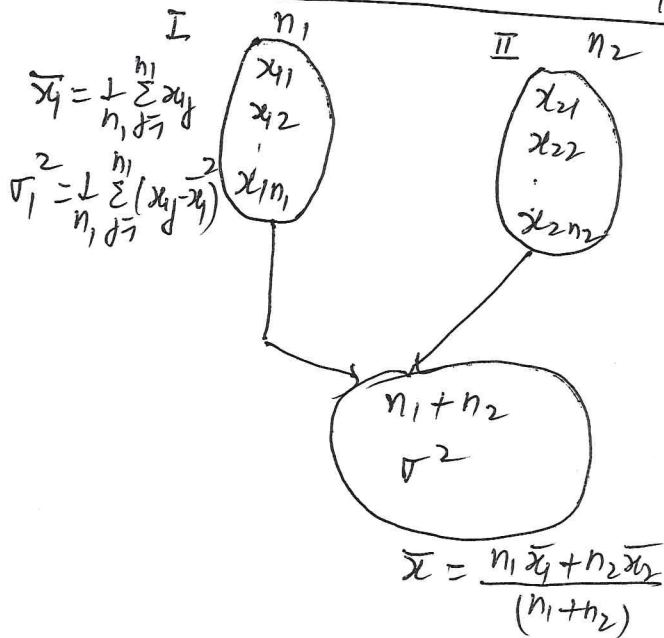
$$\therefore \sigma_x^2 = \frac{1}{N} \sum_{i=1}^n f_i h^2 (U_i - \bar{u})^2$$

$$= \frac{h^2}{N} \sum_{i=1}^n f_i (U_i - \bar{u})^2 = h^2 \sigma_u^2$$

$$\boxed{\sigma_x^2 = h^2 \sigma_u^2}$$

independent of change of origin but depends on scale.

Variance of Combined Series/data



$$\bar{x}_2 = \frac{1}{n_2} \sum_{j=1}^{n_2} x_{2j}$$

$$\sigma_2^2 = \frac{1}{n_2} \sum_{j=1}^{n_2} (x_{2j} - \bar{x}_2)^2$$

$$\sigma^2 = \frac{1}{n_1 + n_2} \left[\sum_{j=1}^{n_1} (x_{1j} - \bar{x})^2 + \sum_{j=1}^{n_2} (x_{2j} - \bar{x})^2 \right]$$

$$= \frac{1}{n_1 + n_2} \left[\sum_{j=1}^{n_1} (x_{1j} - \bar{x}_1 + \bar{x}_1 - \bar{x})^2 + \sum_{j=1}^{n_2} (x_{2j} - \bar{x}_2 + \bar{x}_2 - \bar{x})^2 \right]$$

$$= \frac{1}{n_1 + n_2} \left[\sum_{j=1}^{n_1} (x_{1j} - \bar{x}_1)^2 + n_1 d_1^2 + \sum_{j=1}^{n_2} (x_{2j} - \bar{x}_2)^2 + n_2 d_2^2 \right]$$

$$\text{Where } d_1 = \bar{x}_1 - \bar{x}, d_2 = \bar{x}_2 - \bar{x}$$

$$= \frac{1}{n_1 + n_2} [n_1 \sigma_1^2 + n_1 d_1^2 + n_2 \sigma_2^2 + n_2 d_2^2]$$

$$\sigma^2 = \frac{1}{n_1 + n_2} [n_1 (\sigma_1^2 + d_1^2) + n_2 (\sigma_2^2 + d_2^2)]$$

This proves variance is suitable for further mathematical use.

It can be generalized for K groups of data as

$$\sigma^2 = \frac{1}{\sum_{i=1}^K n_i} \left[\sum_{i=1}^K n_i (\sigma_i^2 + d_i^2) \right]; \text{ where } d_i = x_i - \bar{x}$$

$$(i=1, 2, \dots, K)$$

$$\bar{x} = \frac{\sum_{i=1}^K n_i \bar{x}_i}{\sum_{i=1}^K n_i}$$

Coefficient of Variation (C.V.): Purpose to compare the variability of different sets of data, when their (i) means are not same & (ii) They are measured in different units.

$$C.V. = \frac{S.D}{\text{Mean}} = \frac{\sigma}{\bar{x}}; \text{ In percentage form } C.V. = \frac{\sigma}{\bar{x}} \times 100 = (\%)$$

Ex. Prove that for any discrete distⁿ, standard deviation is not less than mean deviation from mean.

Let

S.D. $\nless$ Mean deviation from mean

$$\Rightarrow (\text{S.D.})^2 \nless (\text{Mean deviation from mean})^2$$

$$(\text{S.D.})^2 \nless (\text{M.D. from mean})^2$$

$$\Rightarrow \frac{1}{N} \sum_{i=1}^n f_i (x_i - \bar{x})^2 \nless \left(\frac{1}{N} \sum_{i=1}^n f_i |x_i - \bar{x}| \right)^2$$

Put $|x_i - \bar{x}| = z$, we have

$$\Rightarrow \frac{1}{N} \sum_{i=1}^n f_i z^2 \nless \left(\frac{1}{N} \sum_{i=1}^n f_i z \right)^2 \nless 0$$

$$\Rightarrow \frac{1}{N} \sum_{i=1}^n f_i z^2 - \left(\frac{1}{N} \sum_{i=1}^n f_i z \right)^2 \nless 0$$

$$\Rightarrow \frac{1}{N} \sum_{i=1}^n f_i (z - \bar{z})^2 \nless 0$$

$\sigma_z^2 \nless 0$ which is always true.

Covariance (Joint variation)

x	x ₁	x ₂	-	-	-	x _n
y	y ₁	y ₂	-	-	-	y _n

$$\text{Cov}(X, Y) = \sigma_{xy} = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

$$= \frac{1}{n} \sum_{i=1}^n x_i y_i - \bar{x} \bar{y}$$

$$= \frac{1}{n} \sum_{i=1}^n x_i y_i - \left(\frac{1}{n} \sum_{i=1}^n x_i \right) \left(\frac{1}{n} \sum_{i=1}^n y_i \right)$$

For X/f ,two types

(I) Raw moments

Moments about origin

Moments about a constant A

(II) Central moments or moments about mean.

Moments about origin⇒ The r th moment about origin is defined as

$$\mu_r' = \frac{1}{N} \sum_{i=1}^n f_i x_i^r$$

$$\mu_0' = \frac{1}{N} \sum_{i=1}^n f_i x_i^0 = 1, \quad \mu_1' = \frac{1}{N} \sum_{i=1}^n f_i x_i = \bar{x} \text{ (A.M.)}$$

$$\mu_2' = \frac{1}{N} \sum_{i=1}^n f_i x_i^2 \dots \text{etc.}$$

⇒ The r th moment about a constant A is defined as

$$\mu_r'(A) = \frac{1}{N} \sum_{i=1}^n f_i (x_i - A)^r$$

$$\mu_0'(A) = \frac{1}{N} \sum_{i=1}^n f_i (x_i - A)^0 = 1$$

$$\mu_1'(A) = \frac{1}{N} \sum_{i=1}^n f_i (x_i - A) = \bar{x} - A, \text{ gives } \underline{\bar{x} = \mu_1'(A) + A}$$

$$\mu_2'(A) = \frac{1}{N} \sum_{i=1}^n f_i (x_i - A)^2 \dots \text{etc.}$$

⇒ The r th central moment (moment about mean) is defined as

$$\mu_r = \frac{1}{N} \sum_{i=1}^n f_i (x_i - \bar{x})^r; \quad \mu_0 = \frac{1}{N} \sum_{i=1}^n f_i (x_i - \bar{x})^0 = 1$$

$$\mu_1 = \frac{1}{N} \sum_{i=1}^n f_i (x_i - \bar{x}) = 0, \quad \mu_2 = \frac{1}{N} \sum_{i=1}^n f_i (x_i - \bar{x})^2 = \sigma^2 \text{ (variance)}$$

$$\mu_3 = \frac{1}{N} \sum_{i=1}^n f_i (x_i - \bar{x})^3, \quad \mu_4 = \frac{1}{N} \sum_{i=1}^n f_i (x_i - \bar{x})^4 \dots \text{etc.}$$

Relation between central moments & moments about origin

$$\begin{aligned} \mu_r' &= \frac{1}{N} \sum_{i=1}^n f_i (x_i - \bar{x})^r = \frac{1}{N} \sum_{i=1}^n f_i (x_i - h_1')^r \\ &= \frac{1}{N} \sum_{i=1}^n f_i \left[\binom{r}{0} x_i^r (h_1')^0 - \binom{r}{1} x_i^{r-1} h_1' + \binom{r}{2} x_i^{r-2} h_1'^2 - \dots \right. \\ &\quad \left. - \dots (-1)^r \binom{r}{r} x_i^0 h_1'^r \right] \end{aligned}$$

$$\begin{aligned} &= \frac{1}{N} \sum_{i=1}^n f_i x_i^r - \binom{r}{1} \frac{1}{N} \sum_{i=1}^n f_i x_i^{r-1} \cdot h_1' + \binom{r}{2} \frac{1}{N} \sum_{i=1}^n f_i x_i^{r-2} h_1'^2 \\ &\quad - \dots (-1)^r \frac{1}{N} \sum_{i=1}^n f_i x_i^0 h_1'^r \end{aligned}$$

gives

$$\mu_r = \mu_r' - \binom{r}{1} \mu_{r-1}' h_1' + \binom{r}{2} \mu_{r-2}' h_1'^2 - \dots (-1)^r h_1'^r$$

$$\mu_2 = \mu_2' - h_1'^2$$

$$\mu_3 = \mu_3' - 3\mu_2' h_1' + 2h_1'^3$$

$$\mu_4 = \mu_4' - 4\mu_3' h_1' + 6\mu_2' h_1'^2 - 3h_1'^4$$

→ Similarly relation between central moments and moments about any point A

$$\begin{aligned} \mu_r &= \frac{1}{N} \sum_{i=1}^n f_i (x_i - \bar{x})^r = \frac{1}{N} \sum_{i=1}^n f_i (x_i - A + A - \bar{x})^r \\ &= \frac{1}{N} \sum_{i=1}^n f_i (d_i - h_1'(A))^r \end{aligned}$$

$$d_i = x_i - A$$

$$h_1'(A) = \frac{1}{N} \sum_{i=1}^n f_i d_i$$

proceed as above we get

$$\mu_r = \mu_r(A) - \binom{r}{1} \mu_{r-1}(A) h_1'(A) + \binom{r}{2} \mu_{r-2}(A) h_1'^2(A) - \dots (-1)^r h_1'^r(A)$$

Effect of change of origin & scale on moments

The r th central moment is written as

$$\mu_r = \frac{1}{N} \sum_{i=1}^n f_i (x_i - \bar{x})^r$$

Consider the transformation $U = \frac{x-A}{h}$;

A & h are suitably chosen constants

$$x_i = A + hU_i \Rightarrow \bar{x} = A + h\bar{U}$$

$$x_i - \bar{x} = h(U_i - \bar{U})$$

$$\mu_r = \frac{1}{N} \sum_{i=1}^n f_i h^r (U_i - \bar{U})^r = h^r \cdot \frac{1}{N} \sum_{i=1}^n f_i (U_i - \bar{U})^r$$

$$\boxed{\mu_r(x) = h^r \cdot \mu_r(u)}$$

independent of origin but depends on change of scale.

For numerical

	C.I	X	f	$U = \frac{x-A}{h}$	fU	fU^2	fU^3	fU^4
Total			N		$\sum fU$	$\sum fU^2$	$\sum fU^3$	$\sum fU^4$

First we calculate raw moments with respect to origin for new variable U

$$\mu'_r(u) = \frac{1}{N} \sum_{i=1}^n f_i U_i^r \text{ gives}$$

$$\mu'_1(u), \mu'_2(u), \mu'_3(u), \mu'_4(u)$$

using these raw moments, we calculate central moments

$$\mu_2(u) = \mu'_2(u) - \mu'^2_1(u), \mu_3(u) = \mu'_3(u) - 3\mu'_1(u)\mu'_2(u) \text{ etc.}$$

finally, we calculate $\mu_r(x) = h^r \mu_r(u)$.

Pearson's β and γ coefficients

$$\beta_1 = \frac{\mu_3}{\mu_2^{3/2}}, \quad \beta_2 = \frac{\mu_4}{\mu_2^2}, \quad \gamma_1 = +\sqrt{\beta}$$

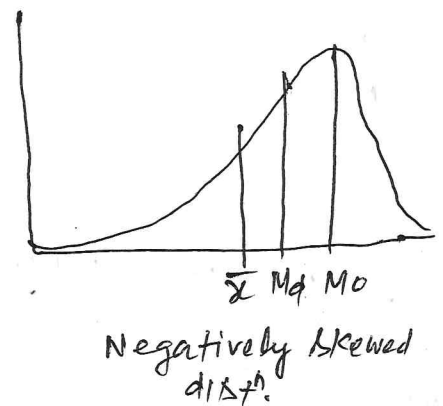
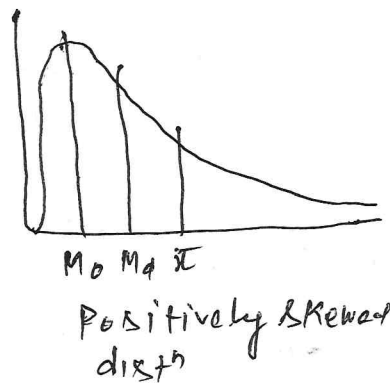
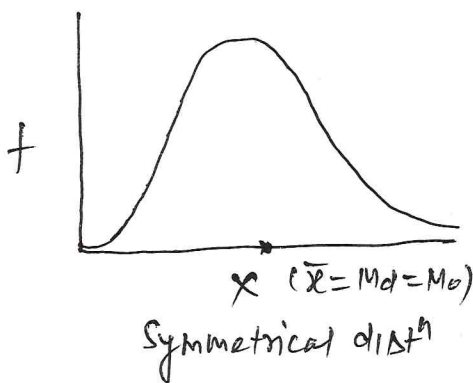
$$\gamma_2 = \beta_2 - 3$$

Skewness

Literally, skewness means lack of symmetry. We study skewness to have an idea about the shape of the frequency curve which we can draw with the help of the given data.

A data is said to be skewed if

- i) Mean $\neq$ Median $\neq$ Mode
- ii) Quartiles are not equidistant from median and
- iii) The ~~freq~~ curve drawn from the data is not symmetrical, stretched more to one side than to other.



Crude measures of skewness

$$i) S_K = \bar{x} - Mo \quad ii) S_K = \bar{x} - Mo$$

$$iii) S_K = (Q_3 - Md) - (Md - Q_1)$$

Prof Karl Pearson's coefficient of skewness

$$S_K = \frac{(\bar{x} - Mo)}{\sigma} = \frac{(\text{Mean} - \text{Mode})}{SD}$$

But we know for moderately asymmetric distⁿ

$$Mo = 3Md - 2\bar{x}$$

then we have

$$S_K = \frac{3(\bar{x} - Md)}{\sigma}$$

$$\boxed{-3 \leq S_K \leq 3}$$

$$\text{as } -1 \leq \frac{\bar{x} - Md}{\sigma} \leq 1$$

Prof. Bowley's coefficient of skewness

$$S_K = \frac{(Q_3 - M_d) - (M_d - Q_1)}{(Q_3 - M_d) + (M_d - Q_1)} = \frac{Q_3 + Q_1 - 2M_d}{Q_3 - Q_1}$$

it is also known as quartile coefficient of skewness

We know if $a > 0$, $b > 0$ $|a-b| \leq |a+b|$

$$\Rightarrow \left| \frac{a-b}{a+b} \right| \leq 1$$

Using this

$$|S_K| \leq 1$$

$$\Rightarrow \boxed{-1 \leq S_K \leq 1}$$

Bowley's & Pearson's coefficient of skewness are not comparable, though for symmetric data both will have $S_K = 0$.

Based upon moments

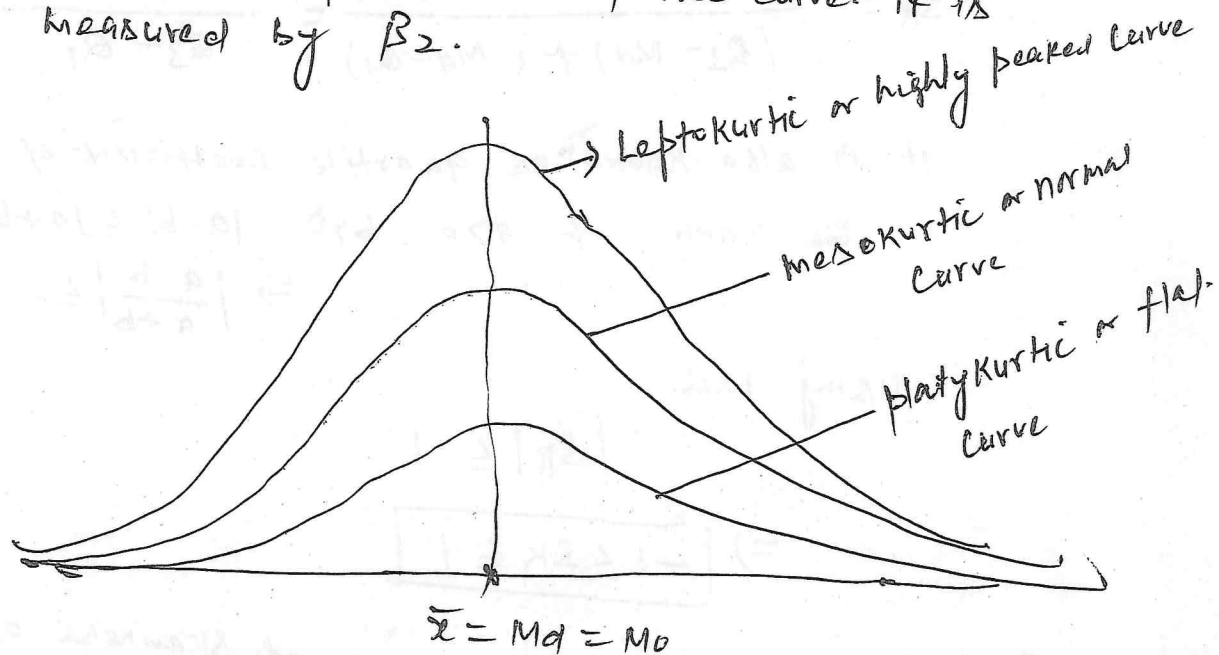
$$S_K = \frac{\sqrt{\beta_1} (\beta_2 + 3)}{2(5\beta_2 - 6\beta_1 - 9)}$$

if $S_K = 0$, either $\beta_1 = 0$ or $\beta_2 + 3 = 0$, but β_2 can't be negative. Hence for symmetric data $\beta_1 = 0$

Hence $\beta_1 = \frac{\mu_3^2}{\mu_2^3}$ can be taken as measure of

skewness. β_1 will only tell whether data is symmetric or skewed. For symmetric $\beta_1 = 0$ for asymmetric data $\beta_1 > 0$. The nature of skewness will be known from μ_3 . For positively skewed data $\mu_3 > 0$ and for -vely skewed data $\mu_3 < 0$, for symmetric data $\mu_3 = 0$.

KURTOSIS: It measures and gives an idea about the flatness or peakedness of the curve. it is measured by β_2 .



For Leptokurtic curve $\beta_2 > 3$ or $\gamma_2 = \beta_2 - 3 > 0$

For Mesokurtic curve $\beta_2 = 3$ or $\gamma_2 = 0$

For Platykurtic curve $\beta_2 < 3$ or $\gamma_2 < 0$

Ex. Show that $\beta_2 \geq 1$

Proof: $\beta_2 = \frac{\mu_4}{\mu_2^2} \geq 1$

$$\Rightarrow \mu_4 \geq \mu_2^2$$

$$\Rightarrow \mu_4 - \mu_2^2 \geq 0$$

$$\Rightarrow \frac{1}{N} \sum_{i=1}^n f_i (x_i - \bar{x})^2 - \left(\frac{1}{N} \sum_{i=1}^n f_i (x_i - \bar{x}) \right)^2 \geq 0$$

Consider $z_i = (x_i - \bar{x})^2$

gives

$$\frac{1}{N} \sum_{i=1}^n f_i z_i^2 - \left(\frac{1}{N} \sum_{i=1}^n f_i z_i \right)^2 \geq 0$$

$$= \frac{1}{N} \sum_{i=1}^n f_i z_i^2 - \bar{z}^2 \geq 0$$

$$= \frac{1}{N} \sum_{i=1}^n f_i (z_i - \bar{z})^2 \geq 0$$

which is always true.

Moments

X	x_1	x_2	-	x_n
f	f_1	f_2	-	f_n

$$N = \sum_{i=1}^n f_i$$

$$\bar{x} = \frac{1}{N} \sum_{i=1}^n f_i x_i$$

I) Raw moments

(a) About A (b) about 0

$$l'_r(A) = \frac{1}{N} \sum_{i=1}^n f_i (x_i - A)^r$$

$$l'_r = \frac{1}{N} \sum_{i=1}^n f_i x_i^r, \quad l'_1 = \bar{x}, \quad l'_0 = 1, \quad l'_2, \quad l'_3, \quad l'_4$$

$$l'_r(A) = \frac{1}{N} \sum_{i=1}^n f_i (x_i - A)^r; \quad \sum_{i=1}^n f_i = N$$

$$= \frac{1}{N} \sum_{i=1}^n f_i d_i^r \quad \text{where } d_i = x_i - A$$

$$l'_r(A) = \frac{1}{N} \sum_{i=1}^n f_i (x_i - A)^r$$

$$= \bar{x} - A \Rightarrow \boxed{\bar{x} = A + l'_1(A)}$$

Relation between moments about mean in terms of moments about any point A and vice-versa

$$l_r = \frac{1}{N} \sum_{i=1}^n f_i (x_i - \bar{x})^r = \frac{1}{N} \sum_{i=1}^n f_i (x_i - A + A - \bar{x})^r$$

$$= \frac{1}{N} \sum_{i=1}^n f_i (d_i - l'_1(A))^r$$

$$= \frac{1}{N} \sum_{i=1}^n f_i (d_i - l'_1(A))^r$$

where $d_i = x_i - A$

$$= \frac{1}{N} \sum_{i=1}^n f_i (d_i^r - r_1 d_i^{r-1} l'_1(A) + r_2 d_i^{r-2} l'^2_1(A) - \dots + (-1)^r l'^r_1(A))$$

$$= l'_r(A) - r_1 l'_{r-1}(A) l'_1(A) + r_2 l'_{r-2}(A) l'^2_1(A) - \dots + (-1)^r l'^r_1(A)$$

$$l_0 = 1, \quad l_1 = 0$$

$$l_2 = l'_2(A) - l'^2_1(A), \quad l_3 = l'_3(A) - 3 l'_1(A) l'^2_1(A) + 2 l'^3_1(A)$$

$$l_4 = l'_4(A) - 4 l'_2(A) l'^2_1(A) + 6 l'_2(A) l'^2_1(A) - 3 l'^4_1(A)$$

Conversely

$$h'_r(A) = \frac{1}{N} \sum_{i=1}^n f_i (x_i - A)^r = \frac{1}{N} \sum_{i=1}^n f_i (x_i - \bar{x} + \bar{x} - A)^r$$

$$= \frac{1}{N} \sum_{i=1}^n f_i (z_i + h'_1(A))^r \quad \text{where } z_i = x_i - \bar{x}$$

$$\bar{x} = A + h'_1(A)$$

$$h'_r(A) = \frac{1}{N} \sum_{i=1}^n f_i (z_i^r + r z_i^{r-1} h'_1(A) + \frac{r(r-1)}{2} z_i^{r-2} h'^2_1(A) + \dots + h'^r_1(A))$$

$$= h_r + r h_{r-1} h'_1(A) + \frac{r(r-1)}{2} h_{r-2} h'^2_1(A) + \dots + h'^r_1(A)$$

In particular, putting $r=2, 3$, and 4 and noting that $h_1 = 0$, we get

$$h'_2(A) = h_2 + h'^2_1(A)$$

$$h'_3(A) = h_3 + 3 h_2 h'_1(A) + h'^3_1(A)$$

$$h'_4(A) = h_4 + 4 h_3 h'_1(A) + 6 h_2 h'^2_1(A) + h'^4_1(A)$$

These formula enable us to find the moments about any point, once the mean and the moments about mean are known.